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Improving Validity and Efficiency of Digital Dyslexia Screening Through Trial-by-Trial Feedback

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Trial-by-trial feedback is a powerful tool across scientific disciplines, offering real-time information that shapes learning, attention, and behavioral adjustment. However, its role in large-scale educational and psychological assessments remains underexplored. This study examines whether providing trial-by-trial performance feedback during a digital dyslexia screener enhances engagement without compromising score validity or interpretability. Engagement is a critical threat to validity in assessments, particularly among young children in digital universal screening contexts. We conducted two large-scale experiments involving 6,332 students from Grades 1 to 12 in Colombia and the United States in which students were randomly assigned to one of two conditions: (a) informative feedback (auditory cues indicating correct/incorrect) or (b) non-informative feedback (neutral sound). The assessment was administered in two formats: an adaptively ordered English version and a randomly ordered Spanish version. We evaluated differences in response behaviors, item response theory model fit, and concurrent validity between conditions. Results indicate that informative feedback significantly increased compliance and reduced disengagement. Assessment scores and item response theory model fit were comparable across conditions. Notably, informative feedback improved the concurrent validity of scores and was associated with shorter completion times. These findings suggest that trial-by-trial feedback can enhance engagement while preserving score interpretability, leading to more efficient and valid digital universal screening assessments.

Public Significance Statement

This study examines the effects of trial-by-trial feedback, using auditory cues to indicate correct or incorrect after each response, on a digital dyslexia screening assessment. Findings suggest that trial-by-trial feedback can enhance children's test engagement without compromising score interpretability.

Keywords: trial-by-trial feedback, dyslexia screening, test validity, test engagement, item response theory

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Student data in the United States are protected by the terms of service, and student data in Colombia are protected by Colombian law. The code used in this analysis is publicly available on the GitHub repository (https://github.com/yeatmanlab/ROAR-Feedback-Experiment-Manuscript; Ma et al., 2026). This study was not preregistered due to the research–practice partnerships with schools. Please note that the data in Study 1 are also analyzed in another article with a different focus: "A Fair Lexical Decision Task for Monolingual and Multilingual Spanish-Speakers" (https://osf.io/preprints/psyarxiv/qfdpb_v1). Please note that a version of this article is available as a preprint at https://osf.io/preprints/psyarxiv/4uag6_v2. This work was presented at the National Council on Measurement in Education Annual Meeting, 2025, where it won the Best Graduate Student E-board Award.

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Wanjing Anya Ma played a lead role in conceptualization, data curation, formal analysis, methodology, software, writing–original draft, and writing–review and editing and a supporting role in project administration. Mia

continued

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Across scientific disciplines, providing trial-by-trial feedback on response correctness is a well-established experimental paradigm for studying how individuals learn and adjust their behaviors. This immediate information guides learning by signaling rewards (Solomon et al., 2015), shaping perceptual processes (Little & Clifford, 2025), supporting cognitive engagement (Hattie & Timperley, 2007), and facilitating both error correction (Anderson, 1982) and strategy adjustment (Haddara & Rahnev, 2022). Although trial-by-trial feedback is effective in learning contexts, its validity and efficiency in large-scale educational and psychological assessments remain underexplored.

Over the past half-century, educational and psychological assessments have undergone a major transition from paper-and-pencil to computer-based formats. This pivotal shift has motivated extensive research evaluating score equivalence (McDonald, 2002; Mead & Drasgow, 1993; Pomplun et al., 2002) and test validity (Boo & Vispoel, 2012; Butcher et al., 2000; Huff & Sireci, 2001), facilitating enhanced scalability and accessibility of assessment programs worldwide. One prominent innovation resulting from this transition is computerized adaptive testing (Chang, 2015; Weiss & Kingsbury, 1984), which optimizes item selection based on previous responses for increased efficiency. Beyond adaptive testing, digital platforms have introduced numerous interactive opportunities, including flexible feedback mechanisms such as "answer-until-correct" formats (Slepkov & Godfrey, 2019), on-demand hints (Bolsinova et al., 2022), and effort monitoring (Wise et al., 2019). Gamified assessments have also emerged, promoting formative assessment and enhancing motivation and engagement during testing (Attali & Arieli-Attali, 2015; Lumsden et al., 2016; Zainuddin et al., 2020).

While digital assessments offer significant advantages in scalability and accessibility, they also introduce challenges related to self-administration, particularly in sustaining engagement in the test to make sure students always give their best effort (Wise & DeMars, 2005). This concern is especially pronounced among young children (Grades K–2) within the universal screening context, where maintaining engagement is critical for obtaining reliable data from all individuals. Thus, mitigating test disengagement and enhancing the score validity of digital universal screening emerge as crucial measurement issues, bridging research and practice in educational and psychological assessment.

One potential solution is trial-by-trial feedback. While widely used in formative contexts to enhance learning (Black & Wiliam, 1998), its application in standardized assessments, particularly rapid screening tools, introduces complex psychometric considerations. Immediate feedback could inadvertently train participants, inflate performance, and violate psychometric assumptions underlying score interpretation (Lord, 2012; Messick, 1995). Prior research on the effects of feedback in various assessments has yielded mixed results (Attali & Arieli-Attali, 2015; Betz, 1977; Hanna, 1976, 1977; Ling et al., 2017; Roos et al., 1997; Vispoel, 1998), highlighting the need for context-specific investigations into its impact on both the validity and efficiency of measurement, especially within the developmental context of universal screening. Thorough evaluations are essential to preserve measurement quality, support valid score interpretations across different assessment formats, and give researchers, clinicians, educators, and families reliable information for making decisions.

Motivated by this need, this article presents a comprehensive and rigorous investigation into the impact of trial-by-trial informative feedback on test validity and efficiency in the context of dyslexia screening. We focus on two single-word recognition measures from the Rapid Online Assessment of Reading (ROAR) suite: ROAR-Word (Ma et al., 2025; Yeatman et al., 2021) and its Spanish counterpart, ROAR-Palabra (Siebert et al., 2024). ROAR-Word has been validated for predicting individually administered reading assessments, and its adaptive version demonstrates improved test efficiency (Barrington et al., 2023; Ma et al., 2025; Yeatman et al., 2021).

ROAR was designed with gamification elements, including an introductory story that positions the test-taker as a wizard searching for a gate through interactions with multiple characters and visual rewards (earning gold coins after every 10 test items). In addition, the task provides trial-by-trial feedback in the form of auditory cues for correct and incorrect responses. All of these aim to enhance student enjoyment and engagement. However, education researchers, practitioners, and clinicians have raised concerns that giving trial-by-trial feedback may negatively impact lower performing students and those with learning differences, potentially compromising the validity and fairness of the assessment in the context of universal screening. This approach to trial-by-trial feedback departs from traditional, nonfeedback standardized assessment practices in schools and therefore warrants an empirical investigation.

To address these questions, we designed an experiment in which students were randomly assigned to one of two conditions: informative trial-by-trial feedback (correct/incorrect sound) or non-informative feedback (neutral sound). Leveraging two large-scale experiments conducted in Colombia and the United States and applying a suite of psychometric methods, we address the following research questions and aim to provide robust empirical evidence regarding the effect of trial-by-trial feedback in testing across developmental age groups.

Research Question 1: Does trial-by-trial feedback influence students' test performance, including response accuracy, response time, total time spent, and disengagement rates?

Research Question 2: Does trial-by-trial feedback affect the fit of the psychometric model for the assessment and its concurrent validity in relation to an individually administered dyslexia screening measure?

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This research contributes to a fundamental understanding of how the immediate feedback mechanisms interact with cognitive processes during digital assessments and offers critical insights for the design of more valid and efficient universal screening tools in educational and clinical contexts worldwide.

Trial-by-Trial Feedback in Psychological Measurement

Trial-by-trial feedback (or immediate feedback) can be broadly categorized into two types: (a) instructional feedback, which provides guidance on problem-solving strategies, and (b) informative feedback, which simply indicates whether a response is correct or incorrect. In formative or classroom assessments (Black & Wiliam, 1998), trial-by-trial feedback, often provided by teachers, is considered crucial to improve student learning. However, its role within standardized testing contexts is more complex due to potential implications for test validity, reliability, and score interpretations, particularly when the goal is to obtain a stable measure of an underlying ability for universal screening.

From a psychometric perspective, providing trial-by-trial feedback can inadvertently train participants during the assessment itself, leading to inflated performance on later trials due to practice effects. Such inflation can undermine the consistency of scores and complicate their interpretation for diagnostic or classification purposes in universal screening. In item response theory (IRT; Lord, 2012), this type of feedback can distort the relationship between an individual's underlying ability and their responses to test items, violating the assumption that ability estimates should remain independent of external influences. Moreover, frequent feedback may induce spurious correlations among item responses within the same test, violating the IRT assumption that item responses are conditionally independent given the individual's ability. Although certain computerized adaptive testing applications can account for prior feedback effects, construct-irrelevant variance (Messick, 1995) remains a significant concern, especially in rapid screening contexts where precise and unbiased measurement is paramount.

The role of immediate feedback in assessments has been debated for decades. Early research in paper-and-pencil and computer-based testing suggested feedback could enhance performance, particularly for higher achieving individuals (Betz, 1977; Hanna, 1976, 1977). However, other studies found no significant effects on accuracy, though sometimes reducing testing time (Attali & Arieli-Attali, 2015; Roos et al., 1997; Vispoel, 1998). More recent work has also shown mixed outcomes, with feedback sometimes improving performance in adaptive tests (Ling et al., 2017) but not consistently affecting accuracy or completion rates in gamified settings (Attali & Arieli-Attali, 2015). These inconsistent findings underscore that the effects of trial-by-trial feedback are context-dependent, highlighting the necessity of targeted investigations, especially in critical areas like universal screening for reading difficulties, where the stakes for accurate identification are high.

Despite the growing body of research, most empirical studies have been limited by small sample sizes, have primarily involved U.S.-based populations, and have insufficiently focused on issues related to test validity in the context of universal screening. Our study addresses these critical gaps by (a) integrating multiple psychometric tools for a comprehensive analysis of feedback effects on measurement validity, (b) employing large, cross-national samples (Colombia and the United States) that facilitate robust test comparisons and enhance generalizability, and (c) including a broad developmental range of participants to offer deeper insights into the implications of immediate feedback and how that might vary across populations and contexts relevant to global screening efforts.

Context of Dyslexia Screening

Dyslexia refers to a reading difficulty characterized by challenges in accurate or fluent word recognition, poor spelling, and decoding skills (Lyon et al., 2003). In the United States, prevalence estimates range from 15% to 20% of the school-age population, depending on the operational cutoff criteria used (Fletcher, 2009; Kuo et al., 2024; Wagner et al., 2020). Identifying and addressing dyslexia have become increasingly important priorities in education and public health systems worldwide. In the United States, by 2025, 49 states have enacted legislation related to dyslexia, with at least 40 states mandating or having implemented universal screening in schools (Burns et al., 2023; The State of Dyslexia, 2025). Similar legislative and policy efforts are emerging globally, with dyslexia screening becoming more prevalent in countries such as Italy (Barbiero et al., 2019) and Brazil (Germano et al., 2017). Dyslexia screening involves using brief, standardized assessments to identify individuals who may be at risk for (or with characteristics of) dyslexia, allowing for early intervention and targeted instructional support (Fletcher et al., 2021). Beyond informing instructional support, dyslexia screening also serves as a criterion for participant selection in specific research studies and shapes educational policy by guiding resource allocation at local and national levels.

Traditional dyslexia screening tools, such as the Dynamic Indicators of Basic Early Literacy Skills Next (University of Oregon, 2018–2020) and the Comprehensive Test of Phonological Processing (Bruno & Walker, 1999), require trained professionals to administer assessments to each student individually in person. More recently, digital screeners such as the ROAR (Ma et al., 2025; Yeatman et al., 2021) offer enhanced scalability and reduced administration time, making them ideal for universal implementation. However, they also introduce challenges related to self-administration, particularly in ensuring student engagement, a concern that is especially pronounced among young children (Grades K–2) who may lack intrinsic motivation for testing. Furthermore, repeated administrations for progress monitoring necessitate tools that are not only accurate but also engaging and efficient. Trial-by-trial informative feedback, especially when integrated with gamification elements, holds the potential to enhance engagement during digital screening, potentially improving the quality of the data collected without compromising its validity. This study directly investigates this potential, examining whether and how such feedback affects the accuracy and efficiency of a digital dyslexia screener across diverse populations and age groups.

Method

Transparency and Openness

Student data in the United States are protected by the terms of service, and student data in Colombia are protected by Colombian law. The code used for this analysis is publicly available on the GitHub repository (https://github.com/yeatmanlab/ROAR-Feedback-Experiment-Manuscript; Ma et al., 2026). This study was not

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preregistered. Data were analyzed using R, Version 4.4.0 (R Core Team, 2024), and the package ggplot, Version 3.5.1 (Wickham et al., 2023).

Research Design

We conducted two large-scale experiments involving 6,332 students from Grades 1 to 12 in Colombia and the United States, approved by the institutional review board. Students were randomly assigned to one of two conditions: (a) informative feedback (auditory cues indicating correct or incorrect responses) or (b) non-informative feedback (neutral sounds). The assigned condition remained consistent throughout the assessment. Randomization was implemented in the browser: When a student started the test, the app randomly selected a feedback condition for that session. If a student closed the tab and restarted the test, they might receive the same or different condition upon reentry, but they needed to start from the beginning. For each student, only the first complete test was included in the analysis.

The assessment was administered in two formats: a randomly ordered Spanish version (ROAR-Palabra) and an adaptively ordered English version (ROAR-Word). A comparison of ROAR-Word and ROAR-Palabra is presented in Table 1, and more details can be found in the following subsection.

In Study 1, $N = 5{,}360$ monolingual Spanish-speaking students from Colombia (Grades 1–11) participated. Within the Colombian education system, elementary education includes Grades 1–5, followed by basic secondary education in Grades 6–9 and upper secondary education in Grades 10–11. Five students who did not complete the ROAR-Palabra assessment (two from the correct/incorrect group, two from the neutral group, and one who attempted both conditions) were excluded from the analysis. Additionally, we excluded eight students from the correct/incorrect group and nine from the neutral feedback condition who spent more than 30 min on the test, as we believe their testing sessions were likely interrupted. The final analysis includes $N = 5{,}338$ students who completed the ROAR-Palabra within the 30-min threshold. A subset of students ($N = 896$) also completed the Woodcock–Muñoz Letter–Word Identification and Word Analysis as the validation assessments.

In Study 2, we recruited U.S. students ($N = 972$, Grades 1–12) from independent schools specializing in children with learning differences. Three participants from the correct/incorrect group and two participants from the neutral group did not complete the assessment and were excluded from the following analysis. Figure 1 shows the grade distributions across the two feedback conditions. Due to an operational update, 758 students completed the version with 84 adaptive items and 16 new items presented in random order, while 209 students completed the version with only the 84 adaptive items.

Instruments

Computer Adaptive English Test: ROAR-Word

ROAR-Word is an English lexical decision task that assesses single-word reading ability. In this task, students are briefly presented with a real word or a pseudoword (i.e., a nonword orthographically similar to a real word) and asked to classify a fixed number of items in a two-alternative forced-choice format. The presentation time of the item is 350 ms, and students have unlimited time to respond through pressing the left arrow key (pseudoword) or the right arrow key (real word) on the keyboard. Initial validation studies from kindergarten and first- and second-grade students revealed an exceptionally high correlation of $r = .97$ between a 20-min version of ROAR-Word and the gold-standard one-on-one oral reading assessment, the Woodcock–Johnson Letter–Word Identification assessment (Yeatman et al., 2021). Follow-up work shows ROAR-Word can be modeled with a one-parameter logistic (1PL) model with a fixed guessing parameter of .5, and the improved 5-min adaptive version (Ma et al., 2025) achieves a high correlation of $r = .89$ in first grade and $r = .73$ in second grade against the individually administered FastBridge earlyReading measure. Recent validation studies have compared ROAR-Word with the Woodcock–Johnson Letter–Word Identification across K–12, demonstrating a consistently strong correlation ($r = .87$; see the technical manual for details: https://roar.stanford.edu/technical/concurrent-validity-swr.html).

In this study, we used an adaptive version of ROAR-Word (for technical details, see Ma et al., 2025), which consists of 84 adaptively selected items from an item pool of 550 items. In addition, each student responded to 16 new items randomly sampled from an uncalibrated item bank. The student received a new item after each five adaptive items. Notably, the adaptive items were precalibrated in previous studies, where all students received the correct/incorrect audio feedback during testing.

The test begins with gamified instructions, followed by five practice trials to familiarize students with the task. The main testing phase includes five short breaks distributed throughout the assessment. Test-takers have unlimited time to respond to each item and can take as much time as needed before proceeding from a break. The estimated completion time is 5–8 min.

Random-Ordered Spanish Test: ROAR-Palabra

ROAR-Palabra (Siebert et al., 2024) is a newly developed Spanish version of ROAR-Word. The test consists of 120 items,

Table 1
Comparison of ROAR-Palabra and ROAR-Word Tests

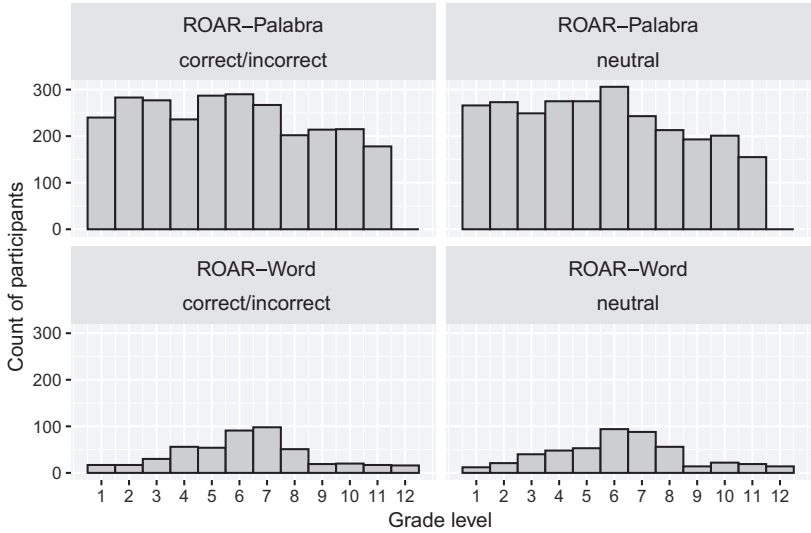
Aessement setting	ROAR-Palabra	ROAR-Word
Language	Spanish	English
Test design	Items are randomly sampled	Computerized adaptive testing
Item pool	70 uncalibrated core items and 308 uncalibrated extended items	550 precalibrated items
Test length	120 items (10–15 min)	100 items (5–8 min)

Note. ROAR = Rapid Online Assessment of Reading.

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Figure 1

Grade Distribution of Participants by the Audio Feedback Conditions ROAR-Palabra and ROAR-Word



Note. The final analysis includes $N = 5{,}338$ students from Colombia and $N = 972$ students from the United States. ROAR = Rapid Online Assessment of Reading.

including both Spanish words and pseudowords presented in a randomized order (for technical details, see Siebert et al., 2024). Each test-taker completes 70 core items and an additional 50 items randomly selected from a larger item pool to enhance measurement precision and minimize item exposure effects. The estimated completion time is 10–15 min. The comparison of ROAR-Word and ROAR-Palabra is presented in Table 1. ROAR-Palabra is not a translation of ROAR-Word. Rather, it was built with both monolingual and multilingual Spanish speakers in mind.

Criterion Measures for Concurrent Validity

To evaluate how trial-by-trial feedback impacts the concurrent validity of ROAR-Palabra, we examined the extent to which its scores align with those from established, individually administered oral reading assessments. Specifically, we administered two subtests from the Woodcock–Muñoz battery (Muñoz-Sandoval et al., 2009): Identificación de letras y palabras (Letter–Word Identification, Woodcock–Muñoz Letter–Word Identification) and Análisis de palabras (Word Analysis, Woodcock–Muñoz Word Analysis). The Woodcock–Muñoz, validated for Spanish speakers in the United States, demonstrates strong internal consistency and reliability between the subtests used in this study (Muñoz-Sandoval et al., 2009; Schrank et al., 2005). These Woodcock–Muñoz reading assessments are often considered the gold standard for evaluating foundational Spanish reading skills (De La Cruz et al., 2019; Khalaf et al., 2019; Mancilla-Martinez & Lesaux, 2011) and can be part of the dyslexia evaluation process. The Woodcock–Muñoz Letter–Word Identification measures a student's oral reading ability, where a student is presented with increasingly difficult Spanish words and is tasked with reading each one out to the proctor. Students are given as much time as they need on each item. The Woodcock–Muñoz Word Analysis assesses a student's decoding abilities by presenting students with pseudowords and asking them to decode each one.

A group of 20 research assistants received extensive training in administration and scoring. A grade-balanced subsample of students who completed ROAR-Palabra was randomly selected to also complete the Woodcock–Muñoz subtests. Students could opt out of this additional testing. Participating students completed the Woodcock–Muñoz battery individually with a proctor in a quiet, distraction-free environment. All sessions were recorded and reviewed for administration fidelity. Furthermore, all scores were independently reviewed by two experienced proctors, and any discrepancies were discussed until consensus was reached.

Analysis

Response Compliance

As a standard practice, previous studies in lexical decision tasks (Ratcliff & McKoon, 2008; Ratcliff et al., 2004) treat responses shorter than 200 ms or longer than 2,000 ms as outliers. In our analysis, responses shorter than 200 ms are classified as rapid guesses, indicating participants did not meaningfully engage with or gather evidence from the test item before responding. Importantly, this differs from the notion of an "educated guess," in which participants typically expend effort and time processing the item but ultimately remain uncertain and must choose arbitrarily. Given that our study is situated within classroom environments involving younger participants who may be more susceptible to distraction, we extended the upper bound of valid responses from 2,000 ms to 5,000 ms. This criterion aligns with previous exclusion criteria (Yeatman et al., 2021).

To examine the relationship between feedback condition and the occurrence of outlier responses, we first examined trends in outlier

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response rates as a function of the number of items administered. Because outlier responses did not occur uniformly across the assessment, the test was divided evenly into three stages: early, middle, and late. For each participant, the mean long response rate and rapid guess rate were calculated within each test stage. To account for small sample sizes in certain grades of ROAR-Word while maintaining consistency in comparisons between the two tasks, participants were grouped into grade-level bins based on their respective educational systems. In Colombia, grade bins were early elementary (Grades 1–2), upper elementary (Grades 3–5), basic secondary (Grades 6–9), and upper secondary (Grades 10–11). In the United States, grade bins were early elementary (Grades 1–2), upper elementary (Grades 3–5), middle school (Grades 6–8), and high school (Grades 9–12). For both ROAR-Palabra and ROAR-Word, Wilcoxon rank-sum tests were conducted to compare long response rates and rapid guess rates between feedback conditions across test stages and grade bins.

To proceed with further analysis, we used previously developed exclusion criteria (Siebert et al., 2024; Yeatman et al., 2024) to identify participants who did not put sufficient effort into the test. This criterion considers both the response time and response accuracy: participants with a median response time less than 450 ms and percent accuracy lower than .65. Supplemental Figure S1 shows that for both the random-order ROAR-Palabra and the adaptive ROAR-Word, high-performing students typically exhibit median response times of at least 450 ms. In contrast, students who respond extremely fast (below 450 ms) rarely achieve accuracy above .65. This pattern suggests a clear speed–accuracy trade-off and provides empirical support that the existing operational threshold works effectively in this data set.

Score Comparability

We followed the same procedure from Siebert et al. (2024) to fit a 1PL model with a fixed guessing parameter of .5 to ROAR-Palabra (see Equation 1). For simplicity, we included the 70 core items from ROAR-Palabra because these items were administered randomly to all participants ($N = 5{,}338$), and we did not apply additional person-level nor response-level exclusions when fitting the IRT model. We used the Expected A Posteriori estimator to obtain ability estimates (θ below) and associated standard errors for each participant. Empirical reliability was calculated using Equation 2. ROAR-Word is an adaptive test that has been in operation for 2 years. It uses an item bank that was precalibrated based solely on responses from the correct/incorrect feedback condition (Ma et al., 2025), using the same IRT model specified in Equation 1. Ability estimates for ROAR-Word were obtained using the maximum likelihood estimation method.

$$\Pr(X_i = 1 \mid \theta) = 0.5 + 0.5\frac{\exp(\theta - b_i)}{1 + \exp(\theta - b_i)}, \tag{1}$$

$$\hat{\rho}_{xx'} = \frac{\widehat{\mathrm{Var}}(\hat{\theta})}{\widehat{\mathrm{Var}}(\hat{\theta}) + \widehat{SE}(\hat{\theta})^2}. \tag{2}$$

Once we excluded participants who did not meet the test compliance criteria, we considered score comparability. Two-sample t tests were performed to compare the means of the scores between the correct/incorrect condition and the neutral feedback condition; these were followed by F tests to examine whether the variance is significantly different. In addition, we fitted and compared generalized linear models to predict theta estimates: One model included only categorical grade-level bin consistent with the previous section (see Equation 3), while the other added feedback condition and its interaction with grade bin (see Equation 4). This analysis aims to provide an additional source of evidence regarding the effect of feedback on score outcomes. Once the effect was identified, further two-sample t tests on score outcomes by grade-level bins were conducted.

$$\hat{\theta} = \beta_0 + \beta_1 \cdot \text{grade} + \epsilon, \tag{3}$$

$$\hat{\theta} = \beta_0 + \beta_1 \cdot \text{audio feedback} + \beta_2 \cdot \text{grade} + \beta_3 \cdot (\text{audio feedback} \times \text{grade}) + \epsilon. \tag{4}$$

We further examined whether the effect of the feedback condition on scores varies across different levels of ability. Inspired by previous work on developing equating relationships between paper-and-pencil and computer-based tests (Pomplun et al., 2002), we equated the scores under two feedback conditions using the equipercentile equating method. Standard errors of equating were calculated and used to develop 95% confidence bands for the equated scores between the two conditions. Although we adopted an equating method, the purpose of our analysis is to investigate the potential score difference across varying ability ranges; therefore, we present our results through a quantile–quantile plot.

Model Fit

Model fitting is a key concern of the test validity. To evaluate if the responses produced by each feedback condition fit the IRT model equally well, we used two approaches. First, we fitted the 1PL model with a fixed .5 guessing separately with the responses from the correct/incorrect condition and with responses from the neutral condition. For each condition, we included all participants and compared item difficulty parameters and item infit and outfit statistics between the two conditions. We used the 70 core items from ROAR-Palabra because these items were administered randomly to all participants. Second, we performed an out-of-sample prediction test (Stenhaug & Domingue, 2022; Watts et al., 2018; Yarkoni & Westfall, 2017) based on the following scheme:

- We split the data into 80% of the training data and 20% of the test data.
- In Group 1, we used the training data to fit an IRT model for each feedback condition and then used the obtained item parameters to calculate the probability of correct for each response in the test data given a participant θ_i and item difficulty b_j. We combined the two sets of probabilities and treated the list of probabilities as the "probabilities from the true feedback."
- In Group 2, we shuffled the training data into two groups and randomly selected half the students to fit one IRT model and the other half to fit the other IRT model. We then used the obtained item parameters to calculate the probability of correct for each response given a participant θ_i and item difficulty b_j. We combined the two sets of probabilities as the "probabilities from the shuffled feedback."

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- We used the InterModel Vigorish (IMV) as an index (Domingue et al., 2024) to compare the predictions between the two groups. If IMV (randomized feedback, true feedback) is closer to 0, it means the effect of feedback on the model fit is ignorable; otherwise, it means the feedback variable plays a role in model prediction ability. It is helpful to note that empirical benchmark values for IMV from PISA mathematics item responses are approximately .01 for the comparison between the 1PL and 2PL models and .002 for the comparison between the 2PL and 3PL models (Domingue et al., 2025).
- We iterated the steps above 10 times to obtain the mean value and standard error of IMV values.

Concurrent Validity

Concurrent validity is a form of criterion-related validity that evaluates a new test's effectiveness by comparing its scores with those of an established measure administered at the same time. To examine if the feedback condition has an effect on the validity of the measurement, we compared ROAR-Palabra with two subtests of the Woodcock–Muñoz (a Spanish parallel of the Woodcock–Johnson IV). Given the fact that test compliance plays an important role in determining the concurrent validity, we included all participants who completed the ROAR-Palabra and had validation data in this analysis.

We used k-fold cross-validation (with $k = 5$) to assess how well scores from ROAR-Palabra between each condition predicted Woodcock–Muñoz scores from a LOESS regression model. The LOESS regression model was chosen because we observed a nonlinear relationship between ROAR-Palabra and the Woodcock–Muñoz scores. In each fold, the data were split into a training set (80%) and a held-out test set (20%), and we computed the Pearson correlation coefficient between predicted and observed scores to evaluate predicted accuracy. This procedure was repeated across all folds, and the average correlation was used as an estimate of the model's concurrent validity.

Process Data

For both ROAR-Word and ROAR-Palabra, we examined differences in participant response behaviors using the process data. After excluding participants who were not test-compliant, we first analyzed the total amount of time participants spent completing an assessment as an indicator of overall test efficiency and engagement. We then decomposed this difference into two components: (a) the median response time for answering individual test items, which provides insight into participants' item processing speed, and (b) the duration of midtest breaks, which reveals participants' test-taking pacing.

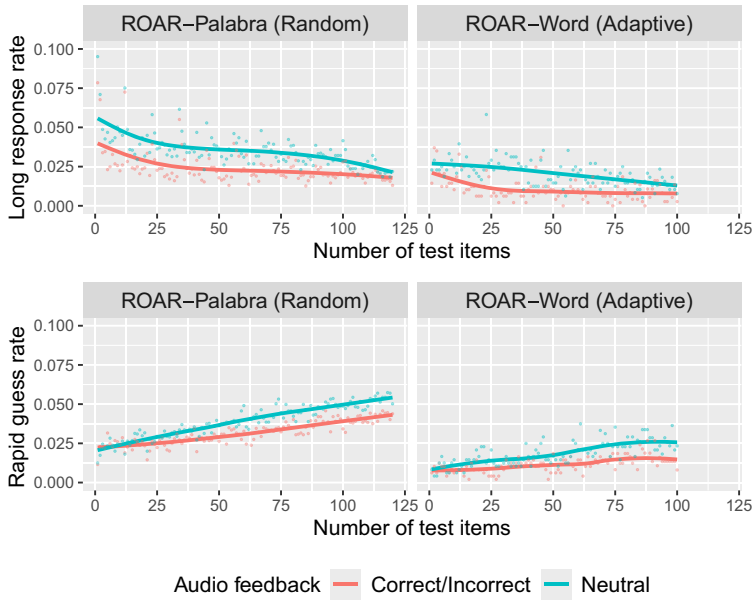
Results

Informative Feedback Leads to Higher Compliance and Fewer Disengaged Participants

The test compliance rate was higher with the correct/incorrect audio feedback than with the neutral audio feedback in both ROAR-Palabra and ROAR-Word. One piece of evidence supporting this finding (see top panels in Figure 2) is the lower rate of long responses (response times exceeding 5,000 ms) in the correct/incorrect condition compared

Figure 2

Test Compliance Analysis Across the Grades



Note. Test compliance is higher with the correct/incorrect audio feedback than the neutral audio feedback in both ROAR-Palabra and ROAR-Word. Long response rate (response time longer than 5,000 ms) decreases and rapid-guessing response rate (response time shorter than 200 ms) increases as students take more items. ROAR = Rapid Online Assessment of Reading. See the online article for the color version of this figure.

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to the neutral condition. This suggests that students were less distracted and maintained greater attention throughout the task when they received informative feedback on a trial-by-trial basis. Moreover, as expected, fewer long responses were observed as students progressed through the test items, indicating that students became increasingly familiar with the task over time. We also observed that students who completed the adaptive version of the task were less likely to produce long responses. This may be explained by improved engagement resulting from task adaptivity. Alternatively, this finding could also reflect greater familiarity with the task or technology among the U.S. student sample, as these students had previously completed other assessments with a similar format.

Additional evidence of test compliance improvement comes from the decrease in rapid-guessing responses (response time shorter than 200 ms) observed in the correct/incorrect condition compared with the neutral condition. Results from both ROAR-Word and ROAR-Palabra (see the bottom panels in Figure 2) suggest that students are more likely to produce rapid guesses in the later stage of the test, and both the correct/incorrect condition and adaptive test mitigate rapid-guessing behavior.

We further analyzed test compliance rates by grade bin and test stage. Table 2 shows that, in the random-order ROAR-Palabra, students receiving the correct/incorrect feedback exhibited significantly fewer outlier long responses across all grade bins in the early and middle test stages, as well as in the late test stage for secondary school students; all effects were small to negligible (ranging from −.15 to −.08). In contrast, significant but negligible effects (−.07) of feedback on rapid guessing were observed only for upper elementary students during the middle and late test stages. These results are consistent with our overall observations in Figure 2 that correct and incorrect feedback could increase participants' engagement; however, the magnitude of this effect was generally small and statistically significant only in certain cases, suggesting a modest improvement in the validity of test scores.

A similar analysis was conducted for ROAR-Word. The feedback condition significantly affected long response rates in upper elementary and high school students, and no significant effects were observed for rapid guessing. It should be noted that ROAR-Word participants were recruited from different schools with varying grade-level distributions. Therefore, the sample is not considered balanced enough to support comparisons across grades, and the results are presented in Supplemental Table S1 for reference.

Using the operational flag system, in which disengaged participants were defined as having a median response time under 450 ms and a percentage correct below 65%, we observed the effect of feedback on engagement. In ROAR-Palabra, we identified 133 out of 2,689 (4.94%) participants in the correct/incorrect feedback condition versus 173 out of 2,649 (6.53%) participants in the neutral condition who were disengaged and randomly guessing. A chi-square test revealed a significant association between the trial-by-trial feedback and the frequency of disengaged participants, $\chi^2(1) = 5.91$, $p < .05$. A similar pattern was observed in ROAR-Word, with 10 out of 486 (2.06%) participants flagged in the correct/incorrect condition and 17 out of 481 (3.53%) participants in the neutral feedback

Table 2

Long Response Rate and Rapid Guess Rate in Rapid Online Assessment of Reading–Palabra: Wilcoxon Rank-Sum Test Results by Grade and Test Stage

Response type	Grade	Test stage	Correct/incorrect (*N*, *Mdn* [IQR])	Neutral (*N*, *Mdn* [IQR])	*W*	*p*	Effect size (*r*)
Long response	Early elementary	Early	523, .05 (.08)	539, .08 (.10)	117697.5	**<.001**[a]	**−.14**
		Middle	523, .05 (.08)	539, .05 (.10)	116393.0	**<.001**[a]	**−.15**
		Late	523, .02 (.08)	539, .05 (.08)	132168.0	.075	−.05
	Upper elementary	Early	800, .02 (.05)	799, .02 (.05)	276593.0	**<.001**[a]	**−.12**
		Middle	800, .00 (.02)	799, .02 (.05)	290570.0	**<.001**	−.08
		Late	800, .00 (.02)	799, .00 (.02)	305378.0	.095	−.04
	Basic secondary	Early	973, .00 (.02)	955, .00 (.02)	413375.0	**<.001**[a]	**−.10**
		Middle	973, .00 (.02)	955, .00 (.02)	407177.0	**<.001**[a]	**−.11**
		Late	973, .00 (.00)	955, .00 (.02)	399204.0	**<.001**[a]	**−.12**
	Upper secondary	Early	393, .00 (.02)	356, .00 (.02)	63397.5	**<.01**	−.08
		Middle	393, .00 (.00)	356, .00 (.02)	59352.5	**<.001**[a]	**−.13**
		Late	393, .00 (.00)	356, .00 (.02)	63128.5	**<.01**	−.08
Rapid guess	Early elementary	Early	523, .05 (.15)	539, .05 (.15)	139092.0	.704	−.01
		Middle	523, .05 (.19)	539, .08 (.18)	133926.5	.154	−.04
		Late	523, .08 (.20)	539, .08 (.25)	136100.5	.326	−.03
	Upper elementary	Early	800, .00 (.00)	799, .00 (.00)	321783.0	.725	.01
		Middle	800, .00 (.00)	799, .00 (.00)	293989.0	**<.001**	−.07
		Late	800, .00 (.00)	799, .00 (.02)	293381.0	**<.001**	−.07
	Basic secondary	Early	973, .00 (.00)	955, .00 (.00)	470167.0	.260	.01
		Middle	973, .00 (.00)	955, .00 (.00)	465703.0	.827	.00
		Late	973, .00 (.00)	955, .00 (.00)	454638.0	.060	−.02
	Upper secondary	Early	393, .00 (.00)	356, .00 (.00)	71266.0	.129	.02
		Middle	393, .00 (.00)	356, .00 (.00)	70032.5	.899	.00
		Late	393, .00 (.00)	356, .00 (.00)	69253.5	.329	−.01

Note. IQR represents the range between the 25th and 75th percentiles. Effect size *r* is calculated as $r = Z/\sqrt{N}$, where *Z* is the standardized test statistic, and *N* is the total number of observations. Significant values appear in bold. IQR = interquartile range.

[a] Test results with both a significant *p* value and a nonnegligible effect size.

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condition; however, this difference did not reach statistical significance, $\chi^2(1) = 1.44$, $p < .23$.

Scores Are Comparable Between Feedback Conditions

Figure 3 shows the distributions of the theta scores in ROAR-Palabra ($N = 5{,}032$) and ROAR-Word ($N = 967$). t test results on score means indicated that there was no statistically significant difference between the feedback groups in both ROAR-Palabra ($p = .807$) and ROAR-Word ($p = .842$). However, an F test indicated a significant difference in variances of the scores between the conditions in both ROAR-Palabra, $F(2475, 2555) = 1.1099$, $p < .01$, and ROAR-Word, $F(463, 475) = 1.2098$, $p < .05$, with higher variance observed in the neutral condition for each task.

In ROAR-Palabra, we found including the feedback condition and its interaction with grade improved the prediction of theta estimates, $\chi^2(4) = 21.398$, $p < .05$. In contrast, in ROAR-Word, adding the feedback condition did not improve score prediction, $\chi^2(4) = 12.984$, $p = .255$. Additional two-sample t tests were conducted to examine if there is any feedback effect on theta estimates by different grade bins, but none of the comparisons showed statistically significant and nonnegligible effects (see Supplemental Table S2).

To further probe the potential difference in performance between two conditions, Figure 4 shows the scale scores for the theta estimates from the neutral feedback group from the correct/incorrect feedback group. In most score ranges, the identity line is inside the 95% confidence band, providing further evidence that the overall scores are comparable. In ROAR-Word (adaptive), we found that lower ability students with theta scores lower than −3 had lower scores in the neutral condition than in the correct/incorrect condition. This result aligns with our interpretation of Finding 1: Informative feedback reduces both rapid guessing and long response times, leading to better performance among lower ability students, who are more prone to these behaviors. We also cannot rule out the possibility that this difference is due to the limited sample size in this score range, as the ROAR-Word sample is relatively small compared with the ROAR-Palabra. In summary, we conclude that scores are comparable between feedback conditions.

In addition, empirical reliability estimates were also comparable across the two feedback conditions. We focus on ROAR-Palabra, which employs a randomly ordered test design that supports fair comparisons of reliability across groups. When item parameters were jointly calibrated using data from both conditions, empirical reliability was high and similar for the correct/incorrect condition (.908) and the neutral condition (.915).

IRT Models Fit Equally Well Between Two Feedback Conditions

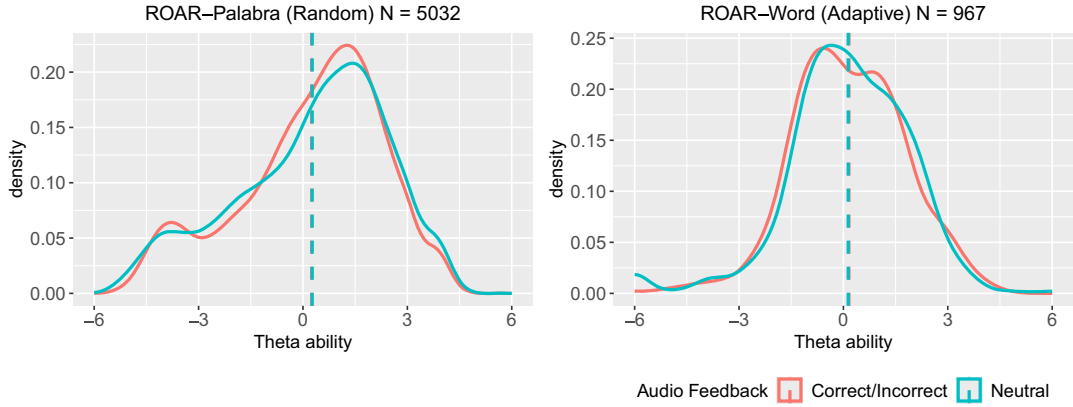
First, item difficulty parameters were highly correlated between the two feedback conditions ($r = .98$; see Figure 5). We also found item infit ($r = .73$) and outfit ($r = .79$) statistics are also correlated between two conditions, although some items from the neutral condition tend to have poorer outfit (see Supplemental Figure S2).

By comparing the predictive performance between models incorporating the true feedback effect and those with a shuffled feedback effect, we found that the IMV value was close to zero ($M = .0002$, $SD = 7.91 \times 10^{-5}$). These results suggest that accounting for feedback group membership does not meaningfully improve model fit. Overall, differences in audio feedback did not appear to impact model fit in this testing context.

Informative Feedback Enhances the Concurrent Validity

We correlated students' individually measured scores on the Woodcock–Muñoz with predicted scores derived from a LOESS regression of ROAR-Palabra theta scores. To assess whether the correlation between ROAR-Palabra and Woodcock–Muñoz differed significantly between the feedback conditions, we conducted a bootstrap analysis with 1,000 resamples. As shown in Figure 6, in real word reading measure, the correlation for the correct/incorrect feedback condition ($N = 394$, $r = .80$) is significantly higher than the correlation for the neutral feedback condition ($N = 402$, $r = .72$) with a 95% bootstrap confidence interval of the difference [.018, .154]. This finding is also consistent for the Woodcock–Muñoz pseudoword reading measure: The correlation for the correct/incorrect feedback condition ($N = 394$, $r = .80$) is significantly higher than the

Figure 3
Score Distributions Between Feedback Conditions

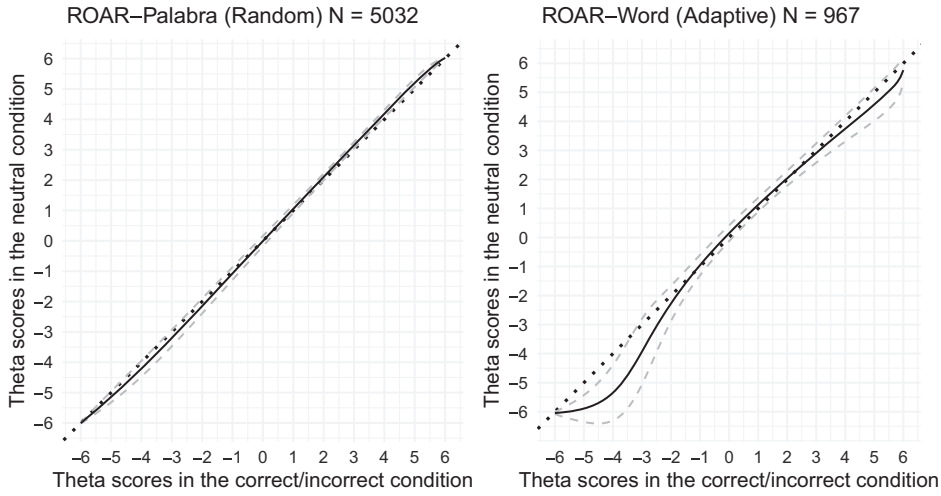


Note. ROAR = Rapid Online Assessment of Reading. See the online article for the color version of this figure.

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Figure 4

The Quantile–Quantile Plot to Compare the Score Distributions Between Feedback Conditions With 95% Confidence Bands



Note. ROAR = Rapid Online Assessment of Reading.

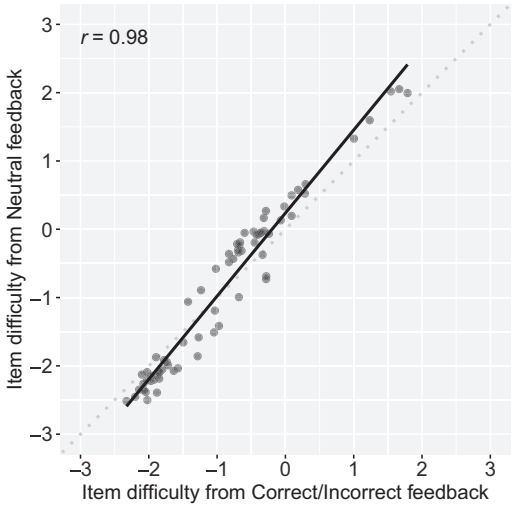
correlation for the neutral feedback condition ($N = 402$, $r = .70$) with a 95% bootstrap confidence interval of the difference [.025, .172]. Results from five-fold cross-validation indicate that theta estimates from the ROAR-Palabra task with informative feedback more accurately predict Woodcock–Muñoz scores than those from the neutral feedback condition (see Table 3).

Informative Feedback Leads to Greater Engagement and Faster Completion of Test

By analyzing the process data shown in Figure 7, we found that students who received the correct/incorrect feedback took significantly less time to complete the test than students who received the neutral feedback. In ROAR-Palabra, total test time was reduced by nearly 15%, and in ROAR-Word, the reduction was approximately 18%. This pattern was consistent across grade levels (see Supplemental Table S3), with larger effects observed among secondary school students in ROAR-Palabra. Further exploratory analyses of item-level response times and time spent during breaks indicated that students who received informative feedback generally responded more quickly at each stage of the assessment than those who received the neutral feedback.

Figure 5

Item Difficulty Comparison Between Two Feedback Conditions



Discussion

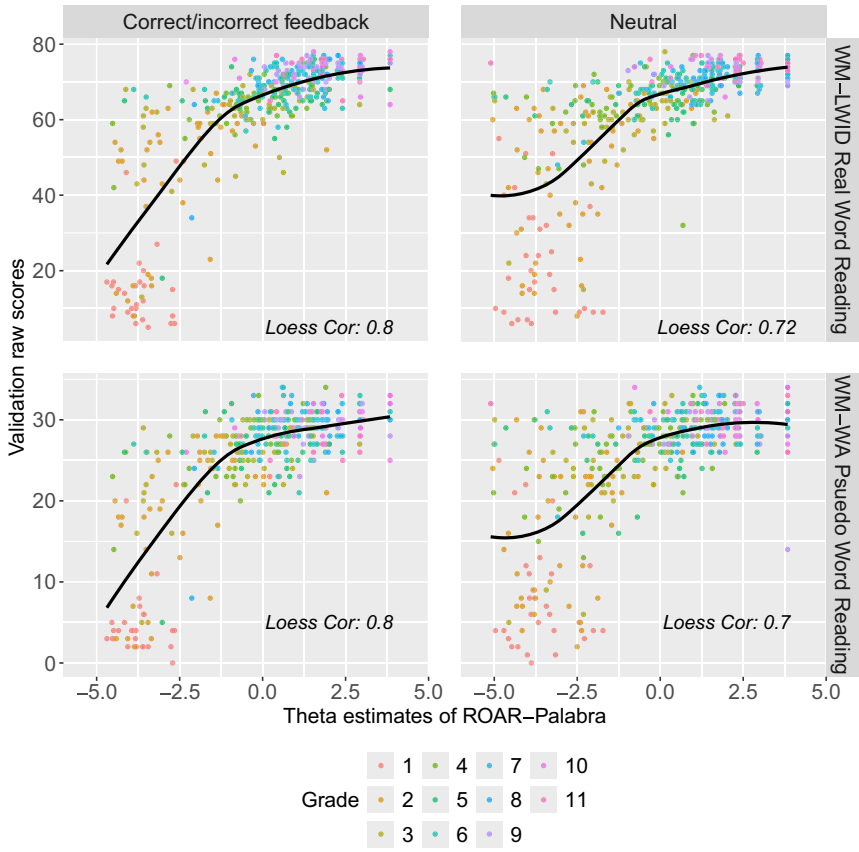
Based on two large-scale experiments involving 6,332 participants across Colombia and the United States, we demonstrated that trial-by-trial feedback enhances engagement without compromising psychometric validity. Given the high prevalence of dyslexia, identifying and addressing it have become increasingly important priorities in both education and public health systems worldwide. In the United States, where more than 40 states have enacted dyslexia screening legislation, efficiently obtaining valid screening results is now an urgent need. Digital assessments hold considerable potential to streamline this resource-intensive process, yet concerns regarding the validity of unproctored digital assessments persist. Our findings indicate that incorporating informative trial-by-trial feedback effectively increases student engagement in test-taking, thereby enhancing both validity and efficiency and offering a more efficient, precise, and interpretable screening measure. Moreover, we believe this approach has the potential to generalize beyond dyslexia screening to other academic and behavioral constructs, advancing universal screening methodologies more broadly (Glover & Albers, 2007).

Identifying careless or rapid-guessing response behaviors to improve test validity is a well-established focus in psychometrics. Common strategies include analyzing response times (Ulitzsch et al., 2023; Wise, 2017; Wise & Kong, 2005) and, more recently, applying mixture models to classify disengaged participants (Uglanova et al., 2025;

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Figure 6

Concurrent Validity of ROAR-Palabra With WM-LWID and WM-WA



Note. All students who completed both ROAR-Palabra and the validation measures were included in the analysis: 394 students in the correct/incorrect feedback condition and 402 students in the neutral feedback condition. Loess Cor measures how well ROAR scores predicted WM scores from a LOESS regression model. ROAR = Rapid Online Assessment of Reading; WM-LWID = Woodcock–Muñoz Letter–Word Identification; WM-WA = Woodcock–Muñoz Word Analysis. See the online article for the color version of this figure.

Ulitzsch et al., 2024; Zhang et al., 2024). However, relatively little attention has been paid to proactive test design strategies that prevent disengaged response behaviors in the first place. In contrast to existing strategies such as effort monitoring paired with proctor notification (Wise et al., 2019) and adjusting test length and sampling frequency (Ulitzsch et al., 2025), our study introduces and validates a lighter touch, trial-level feedback mechanism. This approach not only supports student engagement but also maintains scalability and feasibility, making it particularly well-suited for rapid, large-scale universal screening applications.

To further understand why trial-by-trial feedback increases engagement without inflating performance, we turned to the psychophysics literature, where findings on the effects of such feedback on perception and learning have been mixed (Herzog & Fahle, 1997; Liu et al., 2014; Petrov et al., 2006). In a recent empirical study (Haddara & Rahnev, 2022), researchers found that feedback had no

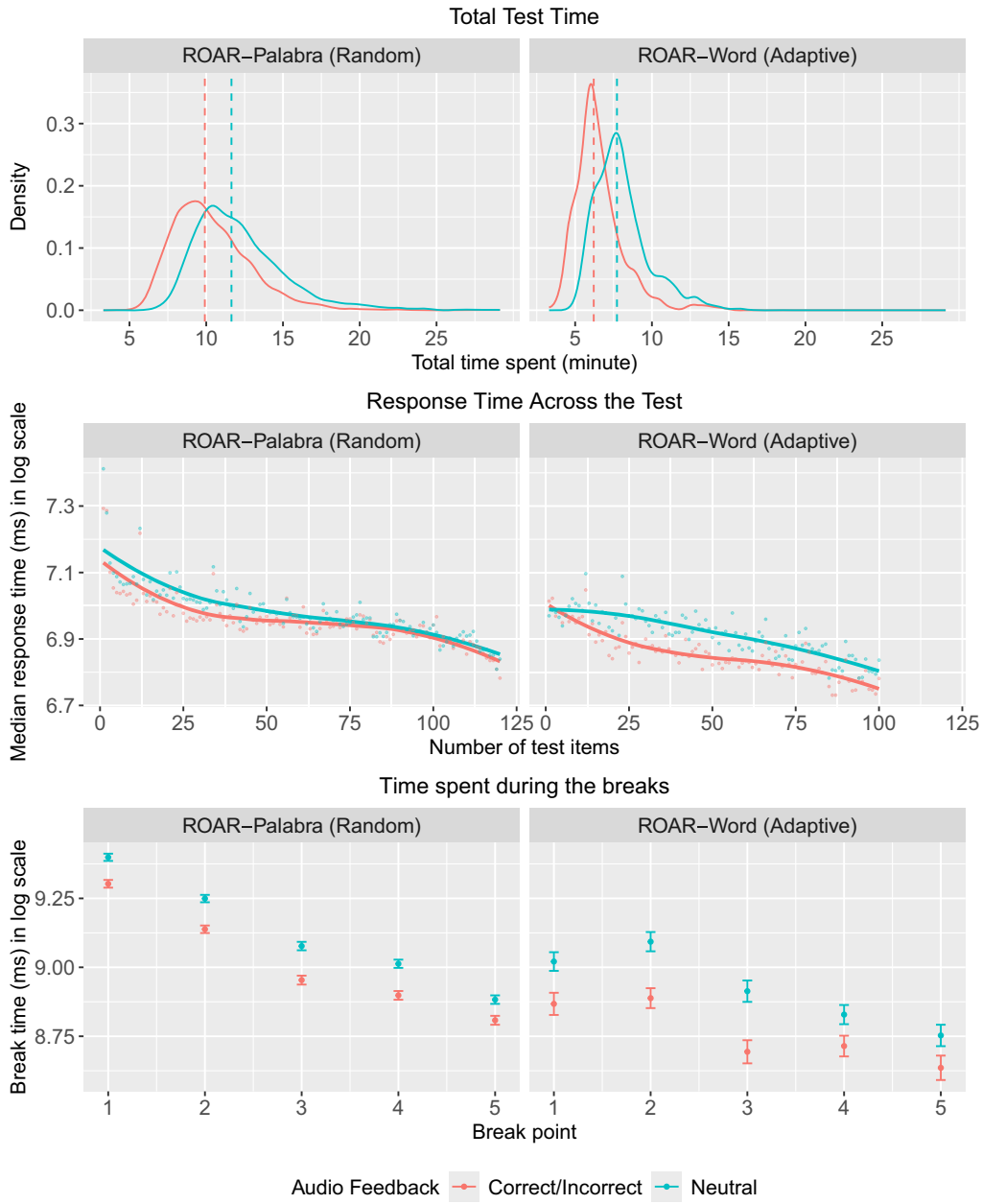
Table 3

Cross-Validated LOESS Performance (Mean ± Standard Deviation) by Predictive Validation Outcome and Feedback Condition

Validation assessment	Feedback	Correlation	Root-mean-square error
Woodcock–Muñoz real word	Correct/incorrect	.794 ± .055	10.102 ± 1.770
	Neutral	.723 ± .059	10.973 ± 1.411
Woodcock–Muñoz pseudoword	Correct/incorrect	.787 ± .052	4.486 ± 0.555
	Neutral	.701 ± .045	5.278 ± 0.648

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Figure 7
Informative Feedback Leading to Faster Completion of Test



Note. ROAR = Rapid Online Assessment of Reading. See the online article for the color version of this figure.

effect on sensitivity but did reduce response biases. They suggested that feedback influences performance through strategy adjustments rather than direct reinforcement. This interpretation may help to explain our results: Students may simply decide to stay engaged in the task as a result of a strategy choice triggered by the informative feedback, rather than showing any actual gain in skill. It is worth noting that previous studies in psychophysics have typically examined learning paradigms, where performance is expected to improve with increased attention and corrective feedback. However, our task assesses single-word recognition, which is a relatively stable latent trait unlikely to improve meaningfully over a short 10- to 15-min session. Thus, we propose that informative feedback primarily enhances attentional engagement as a form of strategy change, rather than reinforcing task proficiency. This type of effect is desirable, as it helps to reveal participants' true underlying ability and thereby improves the psychometric validity of the assessment.

Our results indicate that informative feedback helps maintain students' attentional focus and minimizes distractions, thereby leading to faster test completion. This finding is consistent with previous research involving adult participants (Attali & Arieli-Attali, 2015;

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Roos et al., 1997; Vispoel, 1998); importantly, we extend this test efficiency evidence to a developmental context. Extending previous literature, we specifically addressed test validity such as model fit, score comparability, and concurrent validity, all of which are critical for practitioners interpreting testing results. Notably, through large-scale, rigorous validation procedures, our study illustrates that incorporating informative feedback does not compromise the stability of assessment parameters derived from previous test designs.

Additionally, our study contributes novel developmental insights, as it is among the first to systematically evaluate feedback effects across a normative Grades 1–11 sample in Colombia and a specialized Grades 1–12 sample in the United States. The consistent finding that the distributions of theta scores were equivalent across feedback conditions suggests that providing informative feedback is a broadly applicable strategy that can be implemented across grade levels without compromising score comparability. Beyond score outcomes, in ROAR-Palabra, we found that informative feedback reduced the rate of outlier long responses at all stages of the assessment for secondary school students, but primarily during the early testing stages for elementary school students. We also observed that the effect of informative feedback on reducing total test time was larger among secondary school students than among elementary school students. This pattern suggests that elementary school students may rely on feedback mainly to familiarize themselves with test expectations, whereas secondary school students may use feedback to sustain attention and engagement throughout the assessment, particularly when the test is relatively easy for them.

Despite these strengths, our study has a few limitations. The U.S. sample primarily comprised children with learning differences, contrasting with the more representative general population sample from Colombia. This disparity limits direct cross-country comparisons. However, the consistency of our findings across such diverse populations reinforces the robustness and generalizability of the observed trial-by-trial feedback effects. Another limitation arises from methodological differences: random item ordering in ROAR-Palabra versus computerized adaptive testing in ROAR-Word. The latter showed enhanced test compliance, but confounding factors, such as greater digital familiarity among U.S. students compared to Colombian students, could influence results. In addition, the concurrent validity results would be stronger if validation data comparing ROAR-Word with the Woodcock–Johnson were available. Nonetheless, our two empirical studies provide valuable and complementary insights, raising intriguing questions regarding the generalizability of feedback effects across diverse contexts and settings beyond developed countries and universal screening frameworks. Future research should explore these cross-cultural generalizations and rigorously investigate demographic moderators of trial-by-trial feedback effectiveness to ensure equitable test design across all student populations. In our study, gamification design was controlled and kept consistent across the neutral and correct/incorrect conditions. Future research could use a 2 × 2 factorial randomized controlled test to examine the separate and combined effects of trial-by-trial feedback and gamification on the validity of test scores in digital screening contexts. It would also be valuable for future research to explore alternative formats of trial-by-trial feedback, such as visual or audiovisual feedback.

Finally, our findings underscore the considerable potential of integrating insights and methodologies from adjacent disciplines, such as psychophysics and the learning sciences, into psychological and educational assessment. These findings point to a promising approach for developing assessments that are both personalized and valid (Sinharay et al., 2025). While rigorous validation remains essential, embracing interdisciplinary innovations, even those that challenge traditional model assumptions, may lead to substantial improvements in assessment efficiency and score validity.

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